Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Surds & Indices

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Simplify $\sqrt{50} + \sqrt{18}$. **[3]**

2. Simplify $\sqrt{75} - \sqrt{12}$. **[3]**

3. Simplify $\sqrt{48} + 2\sqrt{27}$. **[3]**

4. Rationalise $\dfrac{1}{\sqrt{5}}$. **[2]**

5. Rationalise $\dfrac{3}{\sqrt{7}}$. **[2]**

6. Rationalise $\dfrac{2}{\sqrt{3}-1}$. **[3]**

7. Rationalise $\dfrac{5}{\sqrt{5}+2}$. **[3]**

8. Rationalise $\dfrac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$. **[4]**

9. Simplify $8^{2/3}$. **[2]**

10. Simplify $27^{2/3}$. **[2]**

11. Simplify $16^{-3/4}$. **[3]**

12. Simplify $\dfrac{4^{3/2}}{2^{-1}}$. **[3]**

13. Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-1)$. **[3]**

14. Expand and simplify $(2\sqrt{5}-1)^2$. **[3]**

15. Simplify $\sqrt{12}\times\sqrt{27}$. **[3]**

Section B – Hard (Q16–25)

16. Rationalise and simplify $\dfrac{3\sqrt{2}+1}{\sqrt{2}-1}$. **[4]**

17. Rationalise and simplify $\dfrac{2+\sqrt{5}}{3-\sqrt{5}}$. **[4]**

18. Solve $x^{2/3}=4$. **[3]**

19. Solve $9^{x}=27^{x-1}$. **[4]**

20. Solve $4^{x+1}=8^{2x-1}$. **[4]**

21. Simplify $\dfrac{\sqrt{18}+\sqrt{8}}{\sqrt{2}}$. **[3]**

22. Express $\sqrt{48}-3\sqrt{12}+2\sqrt{27}$ in the form $a\sqrt{3}$. **[4]**

23. Given $\sqrt{a} + \sqrt{b} = 5$ and $\sqrt{a} - \sqrt{b} = 1$, find a and b. **[5]**

24. Simplify $\left(2^3\right)^{2/3} \times 4^{-1/2}$. **[3]**

25. Solve $\sqrt{2x+1} = x - 1$. **[5]**

End of Worksheet • *Show all working clearly.*